KUSH PATEL

Data Scientist | Applied ML · Forecasting · Experimentation · Causal Inference | Python · SQL · XGBoost · scikit-learn

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Open to remote and relocation across Canada · Legally eligible to work in Canada · Available immediately

SUMMARY

- Data scientist with an MPS in Analytics (Applied Machine Intelligence, GPA 3.76) and 4+ years of paid analytics in supply chain and finance, including SKU-level demand forecasting deployed as Python web apps for a perishable-goods distributor.
- Runs a live wildfire forecast for Canada twice a day and grades every published forecast against what satellites saw afterwards: ROC-AUC 0.734 (95% CI 0.691–0.777) over its first 30 days, ahead of both baselines on ranking, with its top-10% edge reported as not yet distinguishable from zero.
- Experiments and causal inference: uplift models from a ten-arm randomized trial, a proven-optimal targeting plan, and an A/B test of the policy itself against business as usual (+$1.40 per customer, 95% CI $0.23 to $2.57); a geo holdout where six attribution models miss a known answer.
- A model ships only if it beats a simple baseline on data it has never seen, and the losses are published: a learned ranker lost to collaborative filtering (81.2% vs 84.9% recall@10) and a gradient-boosted forecaster lost to a moving average (19.6% vs 18.5% WAPE).

TECHNICAL SKILLS

Modelling: gradient boosting (XGBoost, HistGradientBoosting), logistic and Poisson models, isotonic calibration, anomaly detection (IsolationForest, Mahalanobis distance), recommender systems (item-neighbourhood collaborative filtering, two-stage retrieval and ranking), survival analysis, time-series forecasting

Evaluation: season and out-of-time splits, rolling-origin backtests, ROC-AUC, PR-AUC, top-decile capture and lift, WAPE, recall@k, calibration curves, bootstrap intervals over clustered units, prospective grading of published forecasts, fairness screens

Experiments & Causal Inference: randomized trials and A/B tests, uplift modelling, geo holdouts, difference-in-differences, permutation tests, pre-registered decision rules, power analysis

Optimization & Decisions: mixed-integer programming (HiGHS, Gurobi), LP duals, cost-of-service trade-offs, decision memos with owners and rollback triggers

LLM & NLP: grounded text-to-SQL with retrieval over a semantic layer, ONNX embeddings, Claude API, answer verification against returned rows, on-device speech (faster-whisper, Piper)

Data & Engineering: Python (pandas, NumPy, scikit-learn), SQL (T-SQL, window functions, CTEs), PySpark, Databricks, DuckDB, Microsoft Fabric, dbt, FastAPI, Streamlit, GitHub Actions (scheduled scoring), Docker, pytest, Power BI

PROFESSIONAL EXPERIENCE

Two Rivers Specialty Meats

Dec 2023 – Jun 2026 · North Vancouver, BC

Data Analyst — Operations, Logistics & Enterprise Reporting

- Built SKU-level demand forecasting deployed as Python web apps for procurement, supporting fewer stock-outs and less perishable waste, alongside FEFO and lot-level traceability for recall readiness.
- Delivered customer intelligence for Sales: inactivity signals, lost-account tracking and rep coverage reporting.
- Sole analyst for a perishable-goods distributor: integrated ERP, Square POS, workforce and REST API feeds into curated Bronze/Silver/Gold datasets serving analysis, dashboards and operational reporting.
- Built reconciliation controls comparing source against curated totals for GL/P&L and inventory valuation, cutting discrepancies an estimated ~45% and reporting errors ~30% by fixing issues at the feed (internal figures, directional rather than audited).
- Automated data-quality tests across the mart (schema, completeness, uniqueness, referential integrity), with anomaly thresholds, exception logging and pipeline alerting.
- Engineered ELT in Microsoft Fabric and Azure Data Factory/Synapse using SQL and PySpark, and owned the Power BI semantic model and DAX measure library that Sales, Finance and Supply Chain used.
- Defined governed KPIs with Sales, Finance and Supply Chain (revenue, COGS, gross margin, OTIF, inventory turns, days on hand) with documented definitions and acceptance criteria.

Shivam Investments

Dec 2020 – Jul 2022 · Vadodara, India (hybrid)

Financial Analyst — Data & Reporting

- Automated recurring analysis with Python (pandas) and SQL views feeding Power BI, replacing manual spreadsheet consolidation and cutting manual effort an estimated ~40%.
- Gathered requirements from portfolio and risk stakeholders and delivered Power BI dashboards with auditable measure definitions; performed GL/P&L reconciliations and investigated discrepancies.

SELECTED PROJECTS — public code and CI-tested results; synthetic data is labelled

Canada Wildfire Risk — Python, XGBoost, scikit-learn, GitHub Actions · real public data

github.com/KushPatel29/wildfire-prediction-app · live: wildfire-prediction-app1.streamlit.app

Public forecasts, graded against outcomes: calibrated classifier and Poisson models over 753 cells and 4.0M cell-days of fire and station-weather records. Historical season-split ROC-AUC 0.881; the first 30 prospective days score 0.734 (95% CI 0.691–0.777), ahead of two baselines on ranking while the top-decile edge remains unresolved. Forecasts rebuild twice daily and are graded weekly against satellite detections using whole-day bootstrap intervals. Found and fixed an index-alignment bug that had put 97.7% of fires in the wrong cell.

Corridor Decision Intelligence — Python, DuckDB, PySpark on Databricks, HiGHS / Gurobi, FastAPI · synthetic customers

github.com/KushPatel29/decision-intelligence-platform · live: corridor-decision-intelligence.streamlit.app

Causal ML and decision optimization: uplift models trained on a ten-arm randomized offer trial feed one mixed-integer program over 112,563 customer-offer decisions, solved to proven optimality. The plan captures 75% of the perfect-knowledge value ceiling against 7% for propensity targeting, and the policy was itself randomized against business as usual: +$1.40 per customer (95% CI $0.23 to $2.57). Features reproduced in PySpark on Databricks to within 2.2e-11.

Marketing Attribution & Incrementality — Python, SQL, causal inference · synthetic data with a known answer

github.com/KushPatel29/marketing-attribution-analytics · live: attribution-vs-truth.streamlit.app

Measurement that survives a known truth: six attribution models chase an answer built into the data and none finds it (last-touch gives “direct” a quarter of conversions; its true share is 1.9%). A powered geo holdout does, so the budget plan is calibrated on the experiment rather than on attribution.

Customer Recommendation Engine — scikit-learn, FastAPI, Streamlit · synthetic data

github.com/KushPatel29/Customer-Recommendation-Engine · live: cross-sell-rep-console.streamlit.app

Choosing the simpler model on evidence: a two-stage learned ranker scored 81.2% recall@10 on a warm-customer holdout and plain item-neighbourhood collaborative filtering scored 84.9% (popularity 75.4%), so the simple model ships, and a randomized pilot rather than offline recall decides what it is worth.

Transaction Monitoring — Python, scikit-learn, SQLite, Streamlit · synthetic data

github.com/KushPatel29/aml-transaction-monitoring · live: aml-transaction-monitoring.streamlit.app

Anomaly detection with an audit trail: five explainable rules plus an anomaly model chosen in a bake-off (IsolationForest, Mahalanobis distance, gradient boosting) find all 60 planted cases where a flat $10,000 rule finds 38, while alerting on a third fewer entities.

Ask Your Data — Python, DuckDB, ONNX embeddings, Claude API, Streamlit · synthetic data

github.com/KushPatel29/ask-your-data · live: ask-your-data-kp.streamlit.app

An LLM application that refuses rather than guesses: text-to-SQL over 71 tables in 11 domains, asked by voice or text. A keyless compiler binds questions to a profiled semantic layer; with a key, Claude writes the SQL through the same read-only guard, and its prose is checked against the returned rows. On a 58-question contract: 46 right, 0 wrong, 12 refused.

More at kushpatel29.github.io: an attrition model with no protected attributes and a fairness screen, survival analysis, and revenue and supply-chain analytics in Power BI.

Model lifecycle (project): MLflow training runs and a local SQLite registry preserve six historical wildfire model versions, evaluation artifacts and prospective-monitoring runs; the live forecast is graded against satellite observations. No employer MLOps platform is claimed.

EDUCATION

Northeastern University — Master of Professional Studies in Analytics (Applied Machine Intelligence), GPA 3.76/4.0 · Vancouver, BC — **Sep 2022 – Jun 2024**

Coursework: machine learning, deep learning, NLP, time series, statistical inference.

Gujarat Technological University — Bachelor of Engineering, Computer Science · India — **Aug 2017 – Aug 2021**

AWARDS & CERTIFICATIONS

First prize, 2024 wildfire-prediction hackathon (now a live forecast) · IBM Python for Data Science, AI & Development · Google Data Analytics Professional Certificate · UC Davis SQL for Data Science · Databricks Lakehouse Fundamentals · Power BI Data Modelling with DAX.